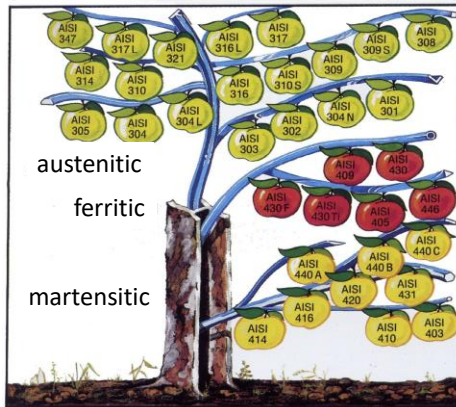




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Technology of Metallic Materials A.Y. 2025-26

Mechanical Engineering Course



Stainless Steels

Giovanni Maizza
DISAT

Lecture 12

1



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Stainless Steels

Excellent corrosion resistance due to high (at least 11.5%) **Chromium** forming chromium oxide (passivating film) over the surface.

Other key elements such as **Ni, Mo and N** may play a leading role in adjusting properties depending on the specific application.

2

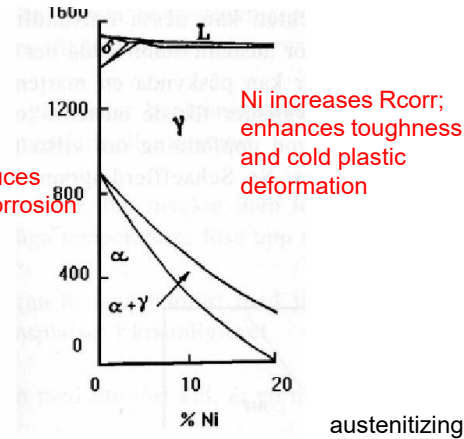
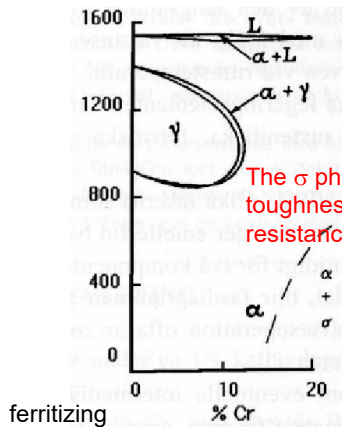


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Phase diagrams: Fe-Cr e Fe-Ni

Cr limits the γ field (completely close); it has ferritizing action, such as Mo, Si, Nb, Ta, Al

Ni widens the γ field (stabilizes austenite) together with C, N, Mn, Cu



3



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CLASSIFICATION OF STAINLESS STEELS

Alloy	C, wt %	Cr, wt %	Ni, wt %	Mo, wt %
FERRITIC	0,010 ÷ 0,12	10,5 ÷ 30	—	0,2 ÷ 2,5
AUSTENITIC	0,015 ÷ 0,15	16 ÷ 28	6 ÷ 32	2 ÷ 7
DUPLEX (austeno-ferritici)	0,030 ÷ 0,050	22 ÷ 28	3,5 ÷ 8	0,1 ÷ 4,5
MARTENSITIC	0,08 ÷ 1,2	11 ÷ 19	—	0,2 ÷ 1,5

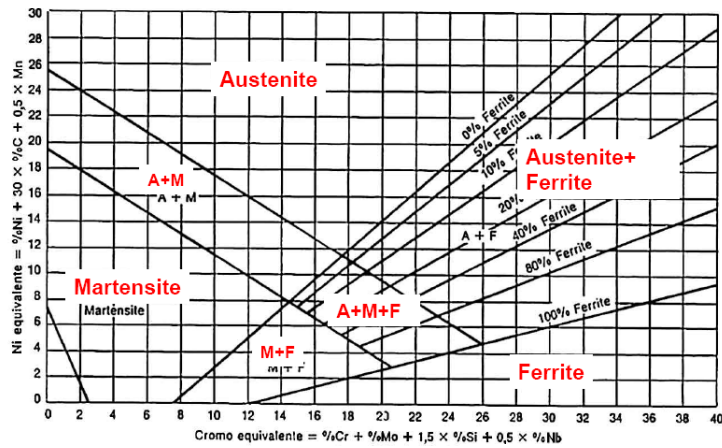
Manufacturing of these alloys requires careful degassing of the liquid since O may oxidize Cr which has high chemical affinity to O thereby losing their corrosion resistance potential.

4



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Schaeffler Diagram



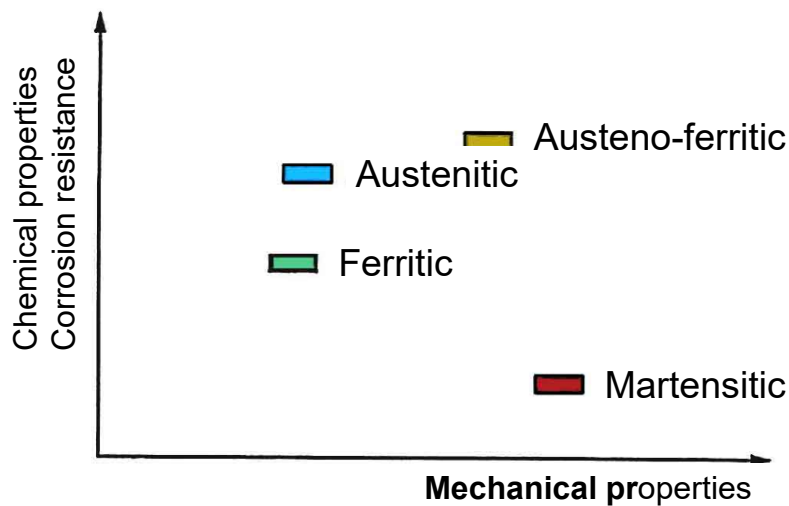
$Cr_{eq} = \%Cr + \%Mo + 1,5\%Si + 0,5\%Nb$ Ferritizing elements
 $Ni_{eq} = \%Ni + 30\%C + 0,5\%Mn$ Austenitizing elements

5



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Chemical vs. Mechanical properties of S.S.



6



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Role of main alloy elements

- Mo** ↑ stability of passivating film especially in reducing environment, ↑ mechanical properties of austenitic especially at high T

- Si** ↑ Rox at high temperature

- Ti, Nb, Ta** carbide-forming elements, refine grains and Impede intergranular corrosion

- Al** As Si, ↑ Rox at high T in refractory steels

- Mn** In austenitic steels, may replace Ni as austenitizing element

7



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AUSTENITIC S.S. (Cr-Ni) – Series 3XX

The presence of Ni expands the austenitic range

Advantages

- Excellent R_{corr}
 - Excellent plastic deformation
 - High toughness
 - Absence of DBTT
 - Good R_{creep}
 - High $R_{fatigue}$
 - Strengthening by cold work
 - Good weldability
 - Paramagnetic properties
- Tensile strength: 559-759 MPa.

Limits

- High price
- Low magnetic properties
- Suffer pitting corrosion

Represents 60%
of the overall production
of S.S.

**Used for chemical equipment,
pressure vessels etc.**

Nb prevents intergranular corrosion
if the alloy is to be used for welding.

8



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FERRITIC S.S. 12-30% Cr

Structure is mainly **ferritic** (BCC α): Cr extends α region and suppresses γ region

Advantages

- Good Rcorr, much higher than Martensitic SS
- higher Rscv than Austenitic SS
- Good plastic deformation (< than Austenitic SS)
- Ferromagnetic props
- < expensive than Austenitic SS
- High strength (517 MPa)

Limits

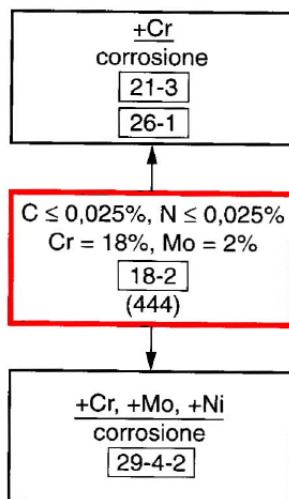
- Generally weldable but tendency to grain growth
- Presence of DBTT

9



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ELI - FERRITIC S.S.



- $\lll \%(\text{C+N}) = 0.02+0.05$
- Rcorr (localized and pitting) higher than common Ferritic SS
- Higher stress corrosion cracking resistance (Rscv)
- Mechanical props similar to conventional Ferritic SS
- DBTT lower than in normal Ferritic

Applications: heat exchangers, vessels, condensers, pumps, food and chemical industry

10



MARTENSITIC S.S.

Martensite formed from **quenching** from austenite region (same as plain carbon steel); when $C > 1\%$, **α loop** is enlarged.

Advantages

- High mechanical strength and hardness
- Ferromagnetic props

Limits

- $< R_{corr}$ than others, espec. in chlorides, possible pitting corr. and stress corr. cracking

Heat treatment: Tensile strength : 517 MPa to 1966 MPa.

Applications: smachine parts, pumps, team turbine blades, valve parts, bolts and screws, cutters, surgery tools, extractive industry, bearings

11



STAINLESS STEELS: weldability

Welding of FERRITIC SS

Possible formation of martensite in melt pool and HAZ.

Risk for grain growth (drop of resilience)

Carbides precipitation and possible formation of sigma phase.

Welding of AUSTENITIC SS

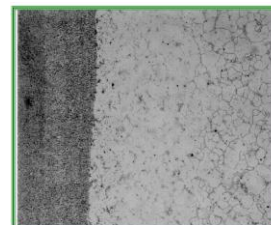
Austenitic SS are most weldable among SS, as they have high initial toughness and good weldability because of the austenitic phase.

However, they may experience precipitation of carbides in the melt pool or HAZ:

-Often precipitation occurs in HAZ because of high CR

In contrast to other carbon steels, SS do not originate cracks after weld cooling.

Austenitic SS



12



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Index of LOCALIZED CORROSION RESISTANCE

- Most of AUSTENITIC SS are susceptible of localized corrosion (e.g. crevice corrosion e pitting corrosion).
- To quantify the corrosion resistance of these alloys a corrosion resistance index has been defined (*Localized Corrosion Resistance Index - LCRI* o *Pitting Resistance Equivalent number - PREN*) :

$$PREN = (A \times \%Cr) + (B \times \%Mo) + (C \times \%N) - (D \times \%Ni)$$

- Where A , B , C , and D are constants.
- According to Nadzhin-Wensley the **PREN** coefficients are:

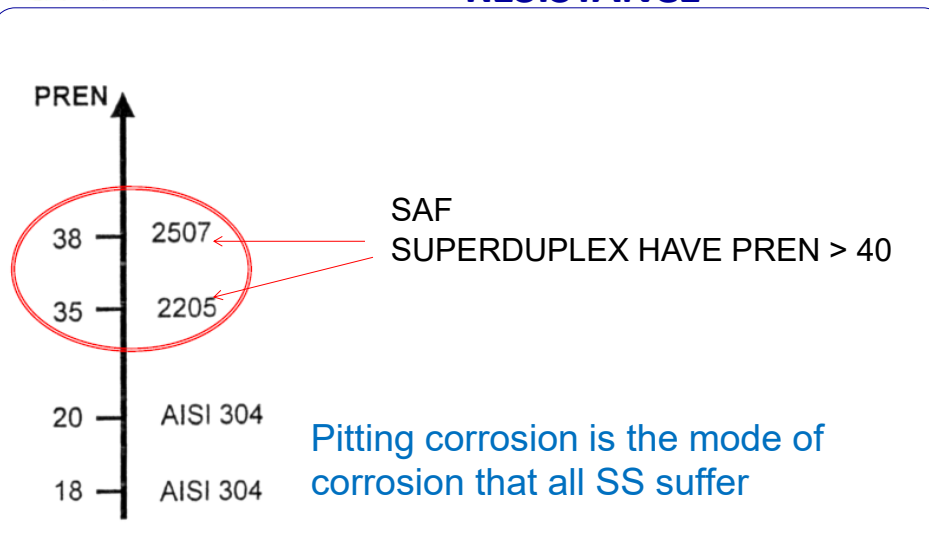
$$PREN = \%Cr + 3,3\%(Mo+0,5\%W) + 16\%N$$

13



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Index of LOCALIZED CORROSION RESISTANCE



14



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Typical S.S. Composition

		C	Cr	Ni	Mo	N	R (MPa)	A (%)
Austenitic	AISI 304	0.06	19	9	-	-	586	55
	AISI 302	0.15	18	9	-	-	620	50
	AISI 316	0.06	17	12	2.5	-	620	50
	AISI 304L	0.03	19	10	-	-	517	55
	AISI 316L	0.03	17	12	2.5	-	517	50
Ferritic	AISI 430	0.12	17	-	-	-	517	30
Martensitic	AISI 410	0.15	12	-	-	-	1310	15
Duplex	23-04	0.02	18	4	-	0.10	700	25
	22-05	0.02	22	5.5	3	0.14	800	25
	25-07	0.02	25	7	4	0.25	920	20
(Precipitation hardening)								
	17-4	0.07	17	4	0.4%Nb			10
	17-7	0.09	17	7	1%Al			6

15



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MECHANICAL PROPERTIES OF SS

S.S.	σ_y (MPa)	%Elong
Ferritic	275-600	20-30
Austenitic	200-380	45-60
Duplex	315-565	20-40
Martensitic	275-1900	15-30
PH	1200-1800	5-15

16



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SUMMARY OF PROPERTIES OF SS

ALLOY GROUP	Magnetic Response (note 1)	Work Hardening Rate	Corrosion Resistance (note 2)	Hardenable	Ductility	High Temperature Resistance	Low Temperature Resistance (note 3)	Weldability
Austenitic	Generally No	Very High	High	By Cold Work	Very High	Very High	Very High	Very High
Duplex	Yes	Medium	Very High	No	Medium	Low	Medium	High
Ferritic	Yes	Medium	Medium	No	Medium	High	Low	Low
Martensitic	Yes	Medium	Medium	Quench & Temper	Low	Low	Low	Low
Precipitation Hardening	Yes	Medium	Medium	Age Hardening	Medium to Low	Low	Low	Low

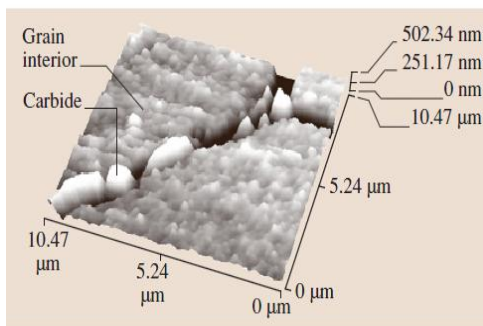
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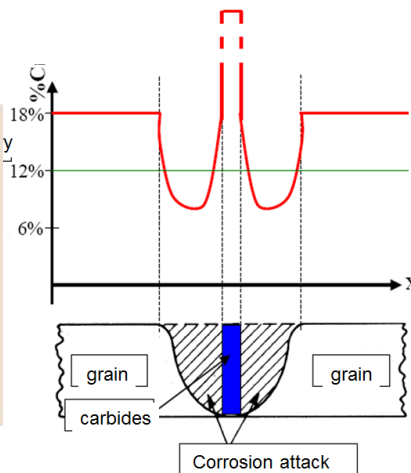
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SENSIBILIZATION OF SS (IN WELDING)

AFM photograph of a highly alloyed stainless steel undergoing intergranular corrosion, leaving behind chromium-rich carbides



Cr-depleted zone

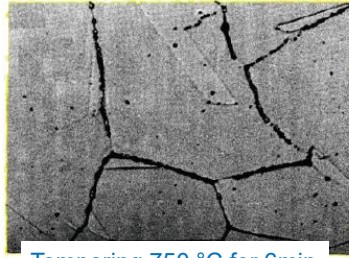


18



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SENSIBILIZATION OF AUSTENITIC SS (AISI 304)



Tempering 750 °C for 6min



Tempering 950 °C for 30 min

Homogenization at 1050 °C

Sensibilization of A.S.S. is generally induced by **welding, cutting, hot plastic deformation** (forging, relaxation heat treatment at high T):

It is caused by the **precipitation of chromium carbides either along the slip bands after plastic deformation or grain boundaries in welding.**

There are a few alloy systems in which very severe attack can occur at grain boundaries, usually as a result of **improper heat treatment.**

19



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SENSIBILIZATION OF AUSTENITIC SS (AISI 304)

Austenitic stainless steels may become **sensitised** to intergranular attack if they are **heated in the range 500-800 °C.**

Chromium carbide precipitates at grain boundaries, and by removing large amounts of chromium is precipitated at grain boundaries, and by removing large amounts of chromium from solid solution renders the zone adjacent to the boundary particularly susceptible to attack.

This effect may be found in the **heat-affected zone adjacent to welds**, if too great a heat input has held the temperature of the zone in the sensitising range for a sufficient period.

The problem is avoided either by **correct heat treatment** or more commonly by **lowering the C < 0.03 %** or stabilising the carbon by the **addition of strong carbide-formers** such as Ti, Ta or Nb so that the carbon is no longer available for chromium carbide precipitation (AISI 316L, AISI 304L)

20



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